Summary

1. HPAC-ML
2. LULESH
    - 2.1 Overview
    - 2.2 Approach 1
    - 2.3 Approach 2

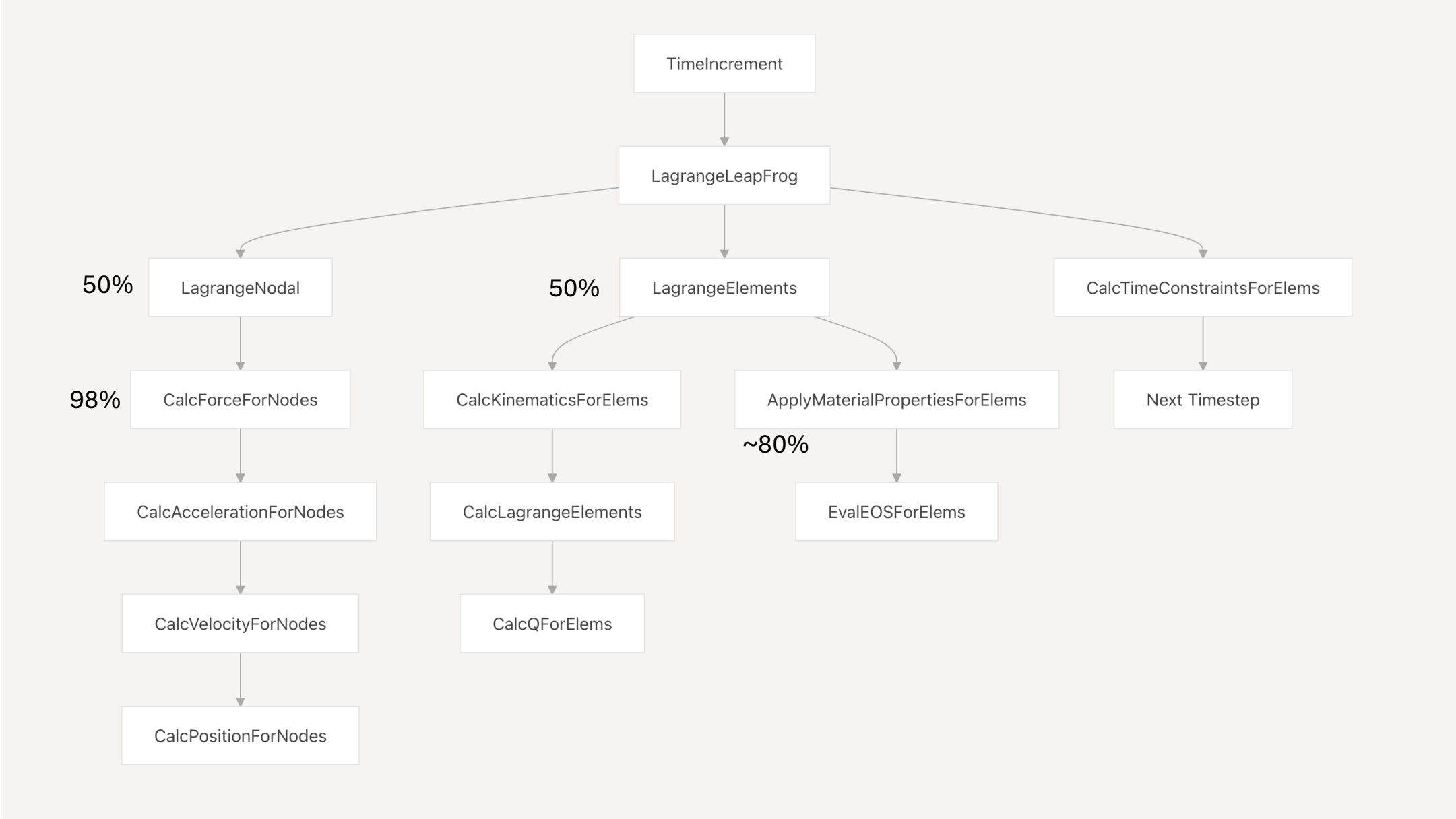
HPAC-ML

- Got it working
- Fixed some issues on the way
  - Part of build process was hardcoded for their GPU versions
- Implemented Feature: support multiple models for inference

LULESH

Livermore Unstructured Lagrangian Explicit Shock Hydrodynamics

- Solves Bast wave problem
- FEM (in spatial dimension) and FDM (in time dimension)
- Lagrangian: Nodes move during simulation
- Hexahedral mesh (in 3D) (i.e. 6-sided, cuboidal)
- Explicit: Integration is explicit


TimeIncrement
LagrangeLeapFrog
50%
LagrangeNodal
50%
LagrangeElements
CalcTimeConstraintsForElems
98%
CalcForceForNodes
CalcKinematicsForElems
ApplyMaterialPropertiesForElems
Next Timestep
~80%
CalcAccelerationForNodes
CalcLagrangeElements
EvalEOSForElems
CalcVelocityForNodes
CalcQForElems
CalcPositionForNodes

LagrangeNodal

- Each element has 8 nodes
- And each node also belongs to 8 elements

Approximation Approach:

- Output: For each node predict 6 floats (position, velocity of node)
- Input: 488 floats

(properties of all relevant connected elements and nodes)

Result:

- Inconsitent result crashes the system (segfault somehow)

```
typedef struct {
  double x;
  double y;
  double z;
  double xd;
  double yd;
  double zd;
  double nodalMass;
} ln_node_input;

typedef struct {
  double q;
  double volo;
  double v;
  double elemMass;
  double ss;
} ln_elem_input;

typedef struct {
  // Main node is connected (belongs to) to 8 elements
  ln_node_input main_node;
  ln_node_input elems_other_nodes[8][7];  // 7 nodes of the 8 elements
  ln_elem_input elems[8]; // 8 elements
} ln_input;

// Output is node properties
typedef struct {
  double x;
  double y;
  double z;
  double xd;
  double yd;
  double zd;
} ln_output;
```

CalcForceForNodes

- For each element it computes some parameters:
  - Element stress = p - q (p is pressure, q is artificial viscosity)
  - Some shape function parameters and normals
  - Using those values, it computes element stresses
- Element stresses are converted to node forces
- Node receives force contribution from all the elements it belongs to
- hourglass control for elements

Input is same as for LagrangeNodal

Output is just the Forces

INITIAL RESULTS

- doesn’t crash (at least for same number of iterations it was trained on)
- Runs 8 times slow (0.11s vs 0.97s)-
- Accuracy: ~